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The Development of Digital Innovator Tri-Ecosystems: A Case Study of a Cryptocurrency Exchange

Research-in-progress

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Abstract

Digital technology is reshaping business and society, driving the emergence of digital innovation as a strategic approach to solving problems and optimising processes through novel applications. This research investigates the development of ecosystems for digital innovators, focusing on the complexities and challenges involved in fostering collaboration among diverse stakeholders. Based on our case study on Uphold, a UK-based cryptocurrency exchange that has survived multiple cryptocurrency bubbles, we have arrived at a perspective of a tri-ecosystem of digital innovator businesses. We also provide several practical implications for digital innovators.

Keywords Crypto-exchanges, Business Ecosystem, Case Study, Digital Innovations

1 Introduction

Digital technology is a fundamental driver of transformation in contemporary business and society (Felicetti et al. 2024; Si et al. 2023). Its widespread impact has spurred digital innovation, defined as the strategic use of digital technologies to solve problems, optimise processes, enhance customer experiences, and create new products, services, and business models (Felicetti et al. 2024; Nambisan et al. 2017). Digital innovation arises from the novel application and recombination of digital components and infrastructure, often demanding significant adaptation from adopters (Ciriello et al. 2018; Fichman et al. 2014). Examples include online retail, the digital transformation of physical stores, ride-hailing platforms disrupting incumbents, and new industries such as unmanned aerial vehicles (Berger et al. 2021). Its distinctive properties—malleability, generativity, and combinatorial potential—enabled by digital platforms and layered modular architectures, differentiate it from traditional innovation and underpin its transformative capacity (Wang 2021). Digital innovators are central to this process, realising ideas, driving organisational change, and navigating the complexities of digital environments to launch ventures, improve public sector performance, and deliver enhanced services (Mancha and Shankaranarayanan 2021; Si et al. 2023; Svahn et al. 2017). Beyond organisational gains, digital innovation enhances competitiveness, fosters adaptability, and offers solutions to societal challenges, including the promotion of digital inclusion (Kolade et al. 2022). Consequently, understanding how the ecosystems that foster these innovations develop is critical for steering them toward the common good. Despite its critical importance, managing digital innovation presents numerous challenges (Brown 2021). Organisations face obstacles: difficulty mobilising resources/partners, and challenges aligning digital initiatives with strategic priorities (Solberg et al. 2020). They also contend with managing competition alongside effective digital innovation and transformation (Joukhadar et al. 2023; Svahn et al. 2017), but traditional innovation theories—such as linear stage-gate models (Cooper 1990) or frameworks focused on physical product architectures (Henderson and Clark 1990)—may not fully capture digital innovation’s nuances, particularly its distributed agency, dynamic processes, and complex interplay between technology affordances/constraints (Nambisan et al. 2017). Digital innovators also require collaboration for success, leading to diverse ecosystem formation (Muthukannan et al. 2021), but differing visions/missions among parties can hinder relationship sustainability.

While the literature has separately examined service, sectoral, and innovation ecosystems (e.g., Jacobides et al. 2018; Lusch and Nambisan 2015), this fragmented view is problematic. It overlooks the core managerial challenge of orchestrating these dimensions simultaneously during development. Consequently, a critical gap remains: we lack a cohesive process model that explains this integration, leaving our understanding of digital innovation strategy incomplete. To address this gap, we ask the following research question: *“How are ecosystems of digital innovators developed?”*

2 Literature Review

2.1 Digital Innovations

Innovation, fueled by proliferating technologies, drives entrepreneurship and business growth (Chen 2014). Digital innovators exhibit entrepreneurial orientation, self-efficacy in entrepreneurship and digital technology, and the ability to leverage digital competencies (Solberg et al. 2020). Researchers examine digital innovations’ characteristics, enablers, and outcomes (see Table 1, with categorisation developed by an inductive and iterative process with concepts compared and grouped based on conceptual similarity, in conjunction with previous literature, such as Tan et al. 2015).

Aspects	Description
Characteristics	
Flexible development process	Digital innovations are less bound by the temporal structure of the innovation process and feature dynamically interdependent innovation processes and outcomes (Papadonikolaki et al. 2022).
Combination of cross-domain components	Expertise in digital technologies is necessary but insufficient to generate digital innovation (Perschina et al. 2019; Shi and Herniman 2023). Conversely, digital innovations often involve new combinations of digital, physical, and social components to produce products (Hund et al. 2021).
Enablers	

Collaboration	Innovation rarely occurs in isolation (Zahra and Nambisan 2011), but often involves collaboration with other entities, forming different forms of ecosystems (Adner and Kapoor 2010).
Intense competition in the industry	Intense competition and industry declines may stimulate the adoption of innovation to help traditional market players stay competitive (Baker 2012), possibly by traditional market players (Sia et al. 2016).
Strategy	Successful digital innovations involve strategic considerations in a digital context (Bharadwaj et al. 2013; El Sawy et al. 2010). This may involve new approaches to manage platform configurations (Henfridsson and Bygstad 2013) and to engage both internal and external forces (Hund et al. 2021).
Organisational capabilities	Digital innovation requires building digital capabilities that align with logic (Lanzolla et al. 2021), such as agility, ambidexterity, generativity, and dynamic capabilities (Ciriello et al. 2018; Hund et al. 2021).
Coordinating entity	While digital innovations are often the outcomes of collaboration between different parties, a coordinator is often required to sense innovation opportunities, control quality, formulate problems, and convert ideas into technical solutions (Jiang and Xu 2020; Shi and Herniman 2023)
Outcomes	
Novel products and services	Innovation processes often involve creating, adopting, and exploiting unbounded, value-adding novelties (e.g., products, services, processes, or business models) by incorporating digital technology (Hund et al. 2021).
The fall of traditional market players	Digital innovation proliferation may lead to the fall of existing market players despite their dominant positions. Examples include Blockbuster (Chopra and Veeraiyan 2017) and some traditional, physical travel agents (Castillo-Manzano and López-Valpuesta 2010).

Table 1. A List of Selective Arguments Related to Business Ecosystems

Innovation processes often involve various ecosystems. Innovation ecosystems are collaborative arrangements where firms combine offerings into coherent, customer-facing solutions (Adner and Kapoor 2010), often requiring a leader for strategy (Durst and Poutanen 2013) and external complementor relationships (Kim et al. 2017). While often discussed alongside business ecosystems (Kim et al. 2017), digital innovation development must also consider stakeholder tensions (Hund et al. 2021; Svahn et al. 2017), potential unprofitability (Nahrkhalaji et al. 2018), and the distinction between innovation and business ecosystems. Despite commonalities, their distinctiveness is often unclear. As digital innovators engage stakeholders, both ecosystem types can emerge, facilitating value exchange.

2.2 Business Ecosystems

Senyo et al. (2019), focusing on digital contexts, defined a business ecosystem as a collaborative environment of entities co-creating value through digital technologies. A business ecosystem involves **multiple entities**, often with a **focal value proposition**. Connected by digital platforms, multi-sided markets, supply chains, and technology systems (Adner 2017), ecosystem participants' contributions are complements rather than inputs required for value creation (Jacobides et al. 2018). The increasing significance of business ecosystems (Teubner and Stockinger 2020) is crucial for FinTech and other digital innovation developers and businesses, as they heavily depend on partners for success (see Li et al. 2023; Teng and Khong 2021). Creating business ecosystems involves aligning diverse entities with a shared vision, often under the guidance of a **leader** who prioritises fairness and **mutual success**. Unlike other forms of business-to-business relationships, there are usually **no fixed boundaries** for a business ecosystem (Cobben et al. 2022). Along with the formation processes, the interactions among ecosystem entities play a key role in their ability to evolve and innovate (Huang et al. 2017). Gradually, the ecosystem can expand, empowering companies to deliver diverse, innovative, and complex solutions to customers through **collaborative value creation**. Furthermore, ecosystem entities can **enhance customer service** while staying focused on their core activities (Au et al. 2019; Tan et al. 2020). When compared with other business structures, the value exchange and relationships within business ecosystems can be more flexible (e.g., Muthukannan et al. 2021). Previous literature often addresses business, service, and innovation ecosystems collectively. Yet, Vargo and Lusch (2016) define a **service ecosystem** as a self-contained system of resource-integrating actors linked by shared institutional

logic, mutually creating customer value through service exchanges. This definition is widely cited in service research (e.g., Ostrom et al. 2021; Sklyar et al. 2019). Granstrand and Holgersson (2020) describe an **innovation ecosystem** as an evolving ensemble of actors, activities, artifacts, and institutions vital for innovative performance, a widely cited concept (e.g., Shaikh and Randhawa 2022). While these concepts involve entities interacting on shared value, some definitional confusion exists. Gomes et al. (2018), for instance, propose that business ecosystems focus on value capture, while innovation ecosystems prioritize value creation, suggesting a firm's developmental transition. Granstrand and Holgersson (2020) also emphasise the stronger collaborative nature of innovation ecosystems. We identify research gaps, including more precise distinctions between ecosystem types, ecosystem openness, collaboration (Jacobides et al. 2018), and complementor management (Adner 2017; Senyo et al. 2019). Business ecosystem variations are not mutually exclusive; they often share goals, such as promoting innovation for success, but precisely mapping their differences and commonalities is desirable.

3 Research Method

We chose the case research method for its strength in exploring processes in emerging phenomena (Pan and Tan 2011). We established specific case selection criteria for our RQ. First, the chosen digital innovator needed to be highly successful, evidenced by user numbers and external articles. Second, the selected firm should adopt diverse strategies, enabling us to identify more development practices. Based on these criteria, we chose Uphold as our case study target. As a highly successful crypto-exchange, Uphold boasts 6 million users, positive external feedback, and even grew during the 2022 crypto-winter. This success clearly indicated a match to our first selection criterion. Furthermore, our preliminary data displayed Uphold's wide range of strategies and collaboration with several partners. This allowed us to identify a broader range of development practices applicable to digital innovators, thus aligning with our second criterion. We conducted data collection in two waves. The first wave took place from August 2022 to March 2023, followed by a second wave from December 2024 to April 2025. Diverse sources included Uphold's official websites, mobile apps, and platform, third-party online forums, and news articles. As a result of the two-wave data collection, we included over 120 articles, 20 screenshots of Uphold's platform (including mobile apps), and 5 videos in our dataset. Our data analysis, conducted concurrently with data collection, employed a grounded theory approach to develop a theory from case data. Following Strauss and Corbin's (1998) principle of moving from specific observations to abstract categories, we adopted the analytical procedures and data structure format of Gioia et al. (2013). This hybrid approach combined grounded theory's inductive rigor with the Gioia method's transparency. Accordingly, raw data were systematically coded into first-order concepts, abstracted into second-order themes, and integrated into overarching dimensions. Theoretical saturation was achieved when additional data produced no new insights (Eisenhardt 1989; Pan and Tan 2011).

4 Preliminary Findings

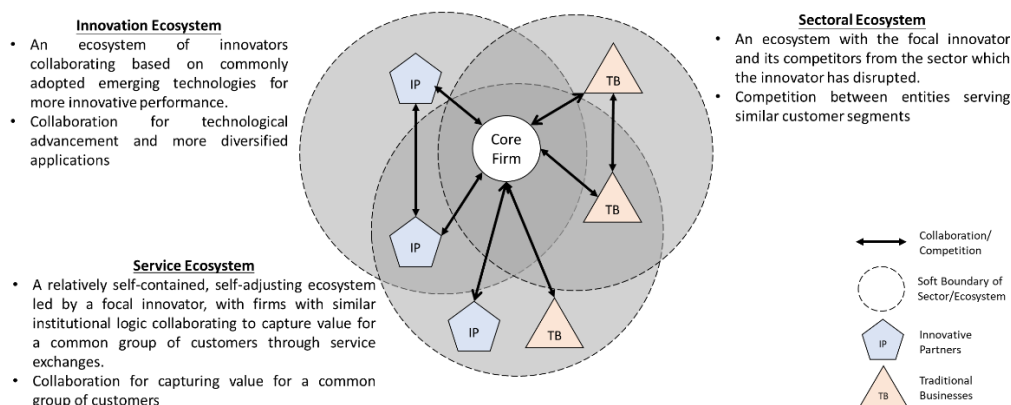


Figure 1: A Tri-Ecosystem of Successful Digital Innovators

Our findings suggest that successful digital innovators, after establishing a minimal structure, gradually assume a significant role within their own service ecosystem, the sectoral ecosystem, and the innovation ecosystem of key emerging technologies, leading to a tri-ecosystem (See Figure 1). These are not three separate, independent ecosystems, but three overlapping and interconnected dimensions of a single, complex strategic environment that the innovator must simultaneously orchestrate.

4.1 Minimal Structure

Inspired by market needs, successful digital innovators started by building a minimal structure, which referred to a lean, technology-centric core with low overhead and minimal physical assets, and was fundamental for business continuity (Au and Ho 2021). In crypto-exchange contexts, investors struggled with the required technical expertise (Fröhlich et al. 2021), creating opportunities for digital innovators like Uphold to establish crypto-exchanges. In this case context, the minimal structure involves realising FinTech critical success factors, such as low-margin, scalability, innovativeness, and transparency (Au et al. 2024; Lee and Teo 2015). For instance, Uphold's API infrastructure enables future innovations with external partners, acknowledged by collaborators. BookingTeam.com's CEO, an online vacation rental platform, highlighted the API's empowering capabilities for their operations.

"Most hosts and property managers are short on time, and often payments and managing money fall last on a long list of to-dos. We choose Uphold as our primary financial services platform because their open API was easy to implement, and their proprietary technology brings speed, ease, and transparency to make our customers' businesses more efficient and effective so they can focus more on delivering top notch service to their guests."

For low-margin, Uphold imposed only a small percentage of the overall transaction amount (typically <1%) with no minimum charge in most scenarios (Chen 2021). This reduced entry service access barriers for unbanked customers (Dierksmeier and Seele 2018). The executive director in India elaborated on their aim to assist unbanked Indians.

"By opening services to India, we continue to move toward our mission to bring the 2.5 billion underbanked accessible and affordable financial service solutions for fast, free, and easy movement of money."

Reaching a transaction volume of USD 500 million in November 2015, Uphold has been selected as the FinTech partner by Antwerp World Diamond. The characteristics cultivated during the early stage would lay the foundation for the development of different ecosystems.

4.2 Service Ecosystem

Once minimally structured, digital innovators must continuously innovate and diversify services to compete and retain customers, as multi-homing or platform switching is increasingly easier for users (Au et al. 2019). However, service providers alone cannot readily achieve this. A practical approach involves forming collaborative relationships with various partners (Masucci et al. 2020; Tan et al. 2020) who share a common value proposition (Adner 2017). This is enabled by instilling variety and innovation at the core of digital innovations. Specifically, Uphold's external partners provided additional methods for users to earn cryptocurrencies, including participating in surveys (via BlockReward.app) and viewing advertisements (using Brave Browser). Uphold also collaborated with Accountintg, Koinly, and TaxBit, three accounting service portals, to facilitate users' cryptocurrency tax reporting.

Consequently, a self-contained, self-adjusting service ecosystem emerged, led by a focal innovator, with firms of similar institutional logic collaborating to capture shared customer value (Vargo and Lusch 2016). Uphold gained competitive advantages and a leadership role within its service ecosystem (Gawer and Cusumano 2014), using user inclination to switch crypto-exchanges due to elevated switching costs from dependence on Uphold's diverse partners (Wang et al. 2016).

4.3 Sectoral Ecosystem

Uphold's scaling gained crypto market influence, though not significantly beyond the early market, necessitating chasm-overcoming efforts (Meade and Rabelo 2004). Consequently, less innovative traditional businesses are engaged as new external stakeholders via partnerships or acquisitions. For example, Uphold acquired Optimus Card to launch debit cards, addressing the need for easier crypto capital withdrawal (Shieh et al. 2023). Uphold's CEO emphasized Optimus's UK Electronic Money Institution (EMI) license's importance in the acquisition announcement.

"We're delighted to have acquired an issuer with such a technically sophisticated platform and a full EMI license... we're very excited to get to work and help scale Optimus's thriving EMD agency business, which already supports several major crypto and fintech ecosystems.....From April, we plan to start rolling out the world's first, multi-asset, crypto-enabled debit card on the Optimus platform in Europe."

Uphold's decision to launch debit cards was welcomed by its users, as indicated by, for example, Internet users' reviews on G2.com.

“The most amazing thing about Uphold is that it offers you a virtual MasterCard for your cryptocurrencies, which is excellent when making online purchases.”

Thus, our findings point to the formation of a sectoral ecosystem. We define this as a system of interdependent actors, including incumbents and challengers, whose relationships are shaped by sector-wide technological shifts and regulation, and that extends beyond any single firm's control. Its value proposition is centered on competition among entities serving similar customer segments. As an authorised card issuer, Uphold could directly compete with traditional financial institutions.

4.4 Innovation Ecosystem

To foster wider blockchain innovation applications, Uphold collaborated with other blockchain innovators, creating an ecosystem based on commonly adopted emerging technologies for more innovative performance (Granstrand and Holgersson 2020) and diversified applications. For instance, Uphold and other technology companies formed the Universal Protocol Alliance (UPA) to facilitate collaborations between blockchain projects and digital asset exchanges. UPA introduced environmental initiatives to tackle cryptocurrency-related issues (Li et al. 2019), including carbon-neutral cryptocurrencies and tradable carbon credit tokens for investment or carbon offsetting. To address challenges and criticisms that might harm public perception of cryptocurrencies and thus impede their proliferation, Uphold consistently advocates for the future and development of cryptocurrencies. For instance, when Elon Musk expressed concerns about Bitcoin's environmental impact, Uphold openly challenged Tesla by launching Bitcoin Zero (BTC0), a carbon-neutral cryptocurrency developed with collaborators in UPA. Their CEO emphasised that BTC0 could be the solution.

“We have the solution. If you come to Uphold and buy some BTC0 and allow Tesla to take payment in the form of this carbon-neutral, rainforest preserving bitcoin, Uphold will buy 5 new Teslas and give them away to those Uphold users who do the most to get this message out there. Rather than waiting for Bitcoin to draw from more sustainable sources for electricity, we've gone out and done something we believe is much better. We've put the highest-quality rainforest preservation derived credits there to mitigate emissions completely.”

These actions were facilitated by Uphold's extensive customer base, without which its customer interactions would be less effective.

5 Discussion and Concluding Remarks

This ongoing study offers several implications. Theoretically, it advances a tri-ecosystem perspective for digital innovators, underpinned by minimal structure and evolving through service, sectoral, and innovation ecosystems. The service ecosystem, consistent with Service-Dominant Logic (Vargo and Lusch 2016), views value co-creation among resource-integrating actors as foundational, adding a temporal dimension by framing this ecosystem as the basis for further development rather than an endpoint. The sectoral ecosystem builds on business ecosystem theory (Gomes et al. 2018), showing digital innovators' evolution from collaborator to disruptor (Adner 2017; Jacobides et al. 2018), shifting from co-creation toward competitive dominance. The innovation ecosystem (Granstrand and Holgersson 2020), centres on sustaining core technologies and addressing “grand challenges,” shaping future technological trajectories. Rather than replacing prior constructs, the tri-ecosystem model integrates them into a developmental framework unique to digital innovators, resolving conceptual ambiguities previously noted (e.g., Jacobides et al. 2018). It shows that these ecosystems are complementary, not competing: the service ecosystem fosters customer retention, the sectoral ecosystem drives competition, and the innovation ecosystem ensures technological resilience. This process-oriented lens provides richer insight than static, single-ecosystem analyses. Practically, the model helps digital innovator managers and founders move beyond product-centric thinking to orchestrate value creation, competition, and innovation resilience. For socially driven digital innovations, it also guides long-term impact and societal benefits. For investors, it offers a framework to evaluate ventures' ecosystem strategies beyond short-term performance. While our study is based on a single case, the model can be analytically generalised (Yin 2009), by comparing with previous literature and other business cases such as Netflix: its service ecosystem integrates studios and ISPs, its innovation ecosystem includes streaming and algorithm developers, and its sectoral ecosystem involves traditional broadcasters and rival streaming platforms (Jenner 2018). This resonance suggests the model's broader relevance and its potential as a foundation for understanding how digital innovators orchestrate multiple ecosystems to ensure survival and sustained value creation. In the future, alternative research methods, such as interviews, may also help identify the boundary conditions and applicability of our theoretical implications.

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